

BHOOMPALLY KALYAN REDDY

ARTIFICIAL INTELLIGENCE & FULL-STACK DEVELOPER

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PROFESSIONAL SUMMARY

Highly motivated Computer Science and Engineering student (2024 – 2028) and published researcher specializing in **Multi-Agent Artificial Intelligence (AI) Orchestration** and **Enterprise Retrieval-Augmented Generation (RAG)**. Demonstrated track record of deploying 30+ products and earning 14 professional certifications. Expert in building autonomous agentic workflows and full-stack platforms with integrated AI capabilities.

SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, SQL, C++, Dart, HTML, CSS, React.js, Next.js, FastAPI, Node.js, Flutter

Artificial Intelligence & ML: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), CrewAI, Natural Language Processing (NLP), Optical Character Recognition (OCR), Computer Vision

Computer Science Foundations: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC), Database Management Systems (DBMS)

Tools & Platforms: Docker, Amazon Web Services (AWS), Git, Ollama, CI/CD, Linux, Cursor AI | **Professional:** Agile (Scrum), Team Leadership

WORK EXPERIENCE

- Founder & Lead AI Developer — AuraCore AI Command Center

January 2026 – Present

[Python](#), [FastAPI](#), [Next.js](#), [CrewAI](#), [Docker](#), [SQLite](#)

 - Spearheaded** the architecture of a multi-agent platform processing **100+ daily queries** with **98% uptime**.
 - Optimized** a high-performance RAG system, reducing retrieval latency by **45%** across 500+ documents.
 - Pioneered** a secure enterprise software licensing system with trial keys, ensuring robust **intellectual property (IP) protection**.
 - Orchestrated** three specialized AI agents using CrewAI to deliver **60% faster** task completion compared to manual workflows.
- AI Research Assistant — IJAR SCT International Journal

March 2026

Research Paper: "Sarathi: Smart Emergency Navigation System" | DOI: [10.48175/IJAR SCT-32733](#) | PDF: [View Paper](#) | Code: [GitHub](#)

 - Co-authored** peer-reviewed research reducing emergency response times by **30%** through intelligent routing algorithms.
 - Formulated** navigation algorithms achieving **0.5-meter proximity accuracy** using real-time GPS coordinates.

PROJECTS

- BKR 2.0: Offline Voice Assistant

Open Source

[Python](#), [Ollama](#), [Vosk](#), [Piper Text-to-Speech](#)

 - Engineered** a private offline voice assistant achieving speech-processing latency under **200 milliseconds**.
 - Enabled** fully offline LLM inference, guaranteeing **100% data privacy** with zero external network dependencies.
- DocuShrink AI: Document Processing Suite

Live Application

[React.js](#), [TypeScript](#), [Tesseract.js](#), [Progressive Web App](#)

 - Delivered** a client-side suite of 10 tools, reducing server bandwidth costs by **100%** via local browser execution.
 - Enhanced** Optical Character Recognition (OCR) capabilities for 13 languages, achieving an **85% character accuracy rate**.
- NutriScan AI: Health and Nutrition Analyzer

Machine Learning

[Python](#), [Jupyter Notebook](#), [Machine Learning](#), [Computer Vision](#)

 - Constructed** an AI-powered computer vision model achieving **92% accuracy** in identifying nutritional content from food images.
- NeuralGuard: Cybersecurity Framework

Security Project

[C++](#), [AI Threat Analysis](#), [Real-time Monitoring](#)

 - Designed** a real-time threat detection engine in C++ analyzing 10+ patterns to identify malware with **95% precision**.

EDUCATION

B.Tech in Computer Science and Engineering

2024 – 2028

ACE Engineering College, Hyderabad, India (Affiliated to JNTUH)

CERTIFICATIONS

IIT Bombay — Python Programming | OutSkill — Generative AI Mastery | Anthropic — Claude AI Professional | IA Online — Technical Training

ACHIEVEMENTS & LANGUAGES

Achievements: Ranked Top 4/200+ in Code Sprint 2026 | Deployed 30+ products | Published Researcher in IJAR SCT | Top 7 in KL University Hackathon

Languages: English (Professional) | Telugu (Native) | Hindi (Conversational)

